On-Chain Evidence, Off-Chain Adjudication: A Layered Approach to Smart-Contract Dispute Resolution

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April 2026

Abstract

This paper makes a focused legal argument about smart-contract dispute resolution. The dominant on-chain dispute-resolution designs—Kleros's Schelling-jury arbitration, Aragon Court's DAO-governed jury, optimistic-rollup challenge windows—attempt to perform adjudication on chain. We argue that this is a category error: settlement is on-chain by nature, adjudication is off-chain by nature, and conflating the two re-introduces the discretionary intermediary that smart contracts were supposed to remove.

We develop the argument by reference to a recently developed settlement primitive (mechanism paper), formally verified in the implementation paper, whose three-layer enforcement architecture explicitly separates the bonding game (Layer 1), the peer-coordination subgame (Layer 2), and the evidentiary record that supports off-chain proceedings when the first two layers fail (Layer 3). The contribution of this paper is to develop Layer 3 in detail: the evidentiary properties of the on-chain record, the relationship between those properties and existing electronic-evidence law (eIDAS, UCC §1-201, the Singapore Electronic Transactions Act, the UNCITRAL Model Law on Electronic Commerce), and the jurisdictional implications of decoupling settlement from adjudication.

The argument is not that smart contracts replace courts. It is that smart contracts can produce a form of evidence that courts already know how to use, and that the cleanest division of labor is one in which the protocol does what it does well (deterministic settlement) and the legal system does what it does well (contextual adjudication). The protocol is silent on what counts as "satisfactory performance" or "material breach"—those are adjudicative determinations—but loud on the facts that adjudication needs as input.

Keywords: electronic evidence, smart-contract law, dispute resolution, blockchain evidence, eIDAS, UNCITRAL, jurisdictional flexibility

1 Introduction

Smart-contract dispute resolution is currently a contested design space. The dominant architectures attempt to perform adjudication on chain, typically through some variant of token-weighted juror selection or DAO governance. Kleros [Kleros, 2019] uses Schelling-point juror voting; Aragon Court [Aragon, 2017] embeds arbitration in DAO infrastructure; optimistic mechanisms [Optimism, 2021, UMA, 2020] use challenge windows in which assertions stand unless contested. Each of these approaches addresses the dispute-resolution problem by importing the discretionary actor that would otherwise reside in a court: the juror, the DAO voter, the challenger.

This paper argues that the import is a mistake. The claim is not that on-chain juries are bad mechanisms (they are interesting mechanisms with documented properties); it is that they are solving the wrong problem. The right problem is not "how do we adjudicate disputes on

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chain" but "how do we produce evidence that off-chain forums can use to adjudicate efficiently." The former requires re-creating, in adversarial token-economic settings, the institutional infrastructure (impartial juror selection, jurisdictional fit, due process) that civil legal systems have refined over centuries. The latter requires only that the on-chain artifact meet existing evidentiary standards—a substantially easier problem, and one for which the legal infrastructure already exists.

We make the argument in the context of FIGARO, a settlement primitive whose mechanism is established in the mechanism paper and whose implementation is verified in the implementation paper. FIGARO explicitly does not attempt on-chain dispute resolution. Its three-layer enforcement architecture treats Layer 1 (bonding) and Layer 2 (peer coordination) as the primary defenses, and Layer 3 (immutable evidence) as the residual mechanism for cases where Layers 1 and 2 are insufficient. The present paper develops Layer 3.

What this paper is not. This paper does not provide legal advice. It does not predict the admissibility of any specific on-chain artifact in any specific jurisdiction, nor does it claim that the on-chain record is itself sufficient for any particular legal claim. It argues for a design discipline (do not perform adjudication on chain) and catalogues the evidentiary properties of the artifact that the discipline produces. Practitioners who wish to rely on those properties in specific proceedings should obtain jurisdiction-specific counsel.

Paper organization. Section 2 states the layered architecture. Section 4 develops the evidentiary properties of the on-chain record. Section 6 compares those properties to existing electronic-evidence law. Section 9 addresses jurisdictional implications. Section 10 argues against on-chain adjudication as a category. Section 12 concludes.

2 The Three-Layer Architecture

FIGARO's enforcement architecture is layered. We summarize it here from the legal-design perspective; the mechanism details are in the mechanism and implementation papers.

Layer 1 — bonding. Each party to a transaction locks $2\times$ the transaction value (asymmetric for N-party processes; the seller's bond scales with upstream value). The locked capital is recoverable only through buyer-initiated atomic resolution. Cooperation is the strictly dominant strategy; defection costs the full bond. The vast majority of disputes are prevented at this layer because the incentive to defect is structurally negative.

Layer 2 — peer coordination. Atomic resolution induces a weakest-link subgame among co-sellers in a process tree: any one seller's defection causes all sellers to lose their bonds. Honest sellers therefore have direct economic incentive to monitor and pressure underperforming co-sellers. This layer handles disputes that survive Layer 1 by internalizing them as intra-cohort coordination problems.

Layer 3 — evidence. The on-chain record. Every commitment, every bond deposit, every resolution (or its absence) is recorded with block timestamps and cryptographic signatures. When Layers 1 and 2 fail—when a party acts irrationally, when a co-seller cohort cannot self-coordinate, when external circumstances (regulatory action, force majeure, participant death) prevent normal resolution—Layer 3 produces the artifact that off-chain dispute forums need to adjudicate.

What is and is not on chain. Settlement is on chain. Adjudication is off chain. The split is deliberate. Settlement is deterministic, automated, and public—all properties that on-chain execution provides natively. Adjudication is contextual, discretionary, and jurisdiction-specific—all properties that on-chain execution is poorly suited for. The split lets each layer do what it does best and avoids forcing one to imitate the other.

3 The Three Layers in Operation

The previous section states the three-layer architecture abstractly. We walk it through operationally here, both because the layered architecture is the paper's central thesis and because practitioners frequently arrive with the question "where does my dispute go?" that the abstract treatment does not directly answer.

3.1 What disagreements look like at Layer 1

Layer 1 disagreements are the ones the bonding rule prevents from happening at all. A buyer who has signed a commitment for a payment P and posted a $2P$ bond faces a choice between cooperation (recover the bond minus the net flow P) and defection (forfeit the bond entirely). The cooperation/defection gap is large in absolute terms (P for the buyer; $P + 2G$ for the seller, where G is the cumulative upstream value); the gap is tight in the technical sense (cooperation strictly dominates defection when the counterparty cooperates, and is equivalent when the counterparty defects). Most participants do not defect, because doing so is locally irrational under the bonded posture.

What does and does not reach a forum at Layer 1. The clearest case for Layer-1 prevention is the *cooperation/defection* dispute: a counterparty who would have walked away from a conventional contract because the cost of performance exceeded the cost of breach is, under the bonded posture, prevented from doing so because breach now costs the forfeited bond. This class of dispute is what the bonding rule forecloses, and it is plausibly the largest class in volume. Real commerce, however, is dominated by a different class: *information-asymmetry* disputes about whether the seller's performance conformed to the agreement (did the goods conform? was time of the essence? did the description match?). The bonded posture pressures the seller toward addressing dissatisfaction, because the seller's bond remains locked while the buyer withholds resolution; but it does not by itself resolve the underlying factual question of whether the buyer's dissatisfaction is justified. Information-asymmetry disputes survive Layer 1 by construction and flow to Layer 2 if multi-party, or to Layer 3 if the parties cannot converge. The buyer-dominance option (do not call `resolveProcess` until dissatisfaction is addressed) is the principal bonded-architecture mechanism for Layer-1 mediation of information-asymmetry disputes; whether it suffices is a function of the dispute's specifics.

3.2 What disagreements look like at Layer 2

Layer 2 disagreements are multi-party coordination disputes that the weakest-link subgame internalizes among the seller cohort. Consider a process tree with a buyer, a primary seller, and three sub-sellers, where one sub-seller is performing poorly enough to threaten the parent process's resolution. Each of the three other sellers (the primary seller and two competent sub-sellers) faces a bond loss of $P_i + 2G_i$ if the underperformer's performance causes the buyer to refuse resolution. They have direct economic incentive to apply pressure on the underperformer: phone calls, replacement-bond financing, escalation through the cohort's coordination channels.

What this looks like operationally. The pressure is endogenous to the bonded structure; no governance, no platform, no aggregator, no foundation organizes it. The result has a formal counterpart in the *Endogenous Coordination Pressure* proposition; from a practical standpoint, the cohort's behavior is its operational form.

Replacement-bond patterns under exploration. Several runtime-tier composition patterns have been proposed for operationalizing Layer 2 mediation but should be understood as exploratory rather than as established mechanisms in the mechanism-paper sense. *Seller substitution* (the buyer commits a replacement order with a new seller before resolution; the underperforming seller's bond forfeits at non-resolution) is the cleanest pattern, since each commitment remains bilateral and bonded. *Assembly-aggregated bonding* (the cohort posts a shared bond pool against cohort performance) shifts coordination from per-seller to per-cohort and raises a distinct set of incentive-compatibility questions the mechanism-paper analysis does not directly address. *Sponsor-of-bond patterns* (a financially-strong party posts bond on behalf of an undercapitalized cohort member) shade into unbonded-actor territory the no-escape-hatches discipline rules out unless the sponsor's own bond is posted alongside, and the sponsor relationship is itself a bonded commitment. Each pattern is a runtime-tier composition above the kernel; the present paper notes them as patterns under exploration and defers full incentive analysis.

3.3 What disagreements look like at Layer 3

Layer 3 disagreements are the residual cases that survive Layers 1 and 2: a participant acts irrationally, the cohort cannot self-coordinate, external circumstances prevent normal resolution, or the underlying agreement is contested on substantive grounds (duress, mistake, frustration, illegality, public policy). The participants take the bonded record to an off-chain forum and the forum adjudicates on the evidence.

The expected rate. The Grameen joint-liability microfinance literature reports headline default-rate reductions on the order of magnitude of 20% → 2% under joint-liability arrangements with peer monitoring [Yunus, 1999]; Morduch [1999] contests the headline figures on accounting grounds (loan rescheduling, definition-of-default ambiguity, selection effects). We do not treat the 2% figure as a peer-reviewed empirical estimate; we treat the 20% → 2% comparison as a stylized indication of the order of magnitude joint-liability peer pressure has been claimed to achieve, recognizing that the headline figure is contested. The bonded primitive's peer-monitoring effect is structurally stronger than Grameen's on three margins (the externality magnitude scales with chain depth; the punishment technology is mechanical rather than social; information availability is on chain), and the on-chain default-and-resolution event log is not subject to the accounting ambiguities Morduch identifies in Grameen's self-reporting. Detailed development of the benchmark is a behavioral-game-theory question; we do not commit to a specific rate prediction.

The forum the parties chose. The forum is named by the parties' off-chain agreement. Layer 3 does not impose a forum; it produces the evidentiary record on which the chosen forum operates. Section 11 below catalogues the forums most commonly composed with the protocol.

3.4 The walkthrough as practical guidance

The three-layer architecture's practical guidance is therefore: expect Layer 1 to handle the vast majority of cases; expect Layer 2 to handle the multi-party residual; expect Layer 3 to handle the small residual that survives both. Plan infrastructure accordingly: most operational deployment work goes into the runtime tier's Layer-1 and Layer-2 surfaces (cart UX, attestation flows, replacement-bond composition, dispute-staging runtime affordances), with Layer 3 being structurally reachable but rarely invoked. A deployment that finds itself routinely escalating disputes to Layer 3 is typically signaling that the upstream layers are mis-configured (insufficient bonding, broken attestation flows, mis-specified clauses), not that the architecture is failing in the way conventional dispute-rich commerce systems would manifest.

4 Evidentiary Properties of the On-Chain Record

Before enumerating the evidentiary properties of the on-chain record, we draw a distinction the remainder of this section relies on. The "on-chain record" is not a single artifact class; it is two artifact classes with substantially different evidentiary character.

Class A: settlement byproduct. Commitment payloads, bond deposits, and resolution calls. These are produced as a mechanical side effect of the bonding game: no party can collect the settlement payout without the corresponding on-chain events having been emitted. Their integrity as evidence is underwritten by the game itself — an actor cannot opt out of producing them without opting out of settlement.

Class B: discretionary attestation. Schema-typed lifecycle events, disclosure records, proximity witnesses, and other party-generated attestations emitted by the attestation coordinator. These are voluntary: a party chooses whether to attest, when to attest, and what content to place in the attestation. The schema validator ensures the content is well-formed; it does not and cannot ensure the content is accurate. A party who selectively attests (silence about underperformance, precise attestation of performance) can produce a Class B record that survives authentication but misrepresents reality.

The six properties enumerated below apply *unconditionally* to Class A and *conditionally* to Class B: each Class B attestation carries the integrity of the signature and timestamp (Class A characteristics), but the accuracy of its content depends on the attestor's incentive structure at the moment of attestation. A cryptoeconomics reader will recognize Class B as subject to the same self-reporting incentive problems that attach to every voluntary disclosure; adjudication should treat Class B under ordinary authentication-and-hearsay doctrine rather than extending Class A's byproduct assurance to it. An attestation-bond layer (stake posted per attestation, slashable on proven falsity) is a plausible Class B strengthening but lies outside the kernel and is not yet developed in the protocol.

Evidentiary completeness and constitutive primacy. A further distinction matters for certain downstream uses: the on-chain record is not merely *admissible* as evidence but *constitutive* of the accounting record of the process. A Figaro process is a self-closing ledger period in the precise accounting sense — the custodial balance of the contract is the balance-sheet position; commit and resolve events are journal and closing entries; the process closes when resolution discharges all custodial liabilities. For disputes whose subject matter is the accounting record of economic activity settled through the protocol, the on-chain record is not corroborating evidence for a party-held book; the chain is the book, and any party-held document is a summary or translation of it. The admissibility analysis of the present section applies to the record's evidentiary weight in proceedings; the constitutive analysis applies to the record's primacy within the economic activity it documents.

The on-chain record produced by Figaro has six properties that together constitute its evidentiary value.

(1) Timestamped. Every state transition is associated with a block timestamp. The timestamp is set by block proposers under consensus rules; it is not subject to retroactive modification. For practical purposes, the timestamp on a finalized block is a qualified time stamp in the eIDAS sense.

(2) Cryptographically signed. Every commitment is signed by both parties using EIP-712 typed structured data [EIP-712, 2017]. The signatures bind the commitment content (payment, parties, denomination, agreement hash, deadline) to the signers' private keys. Recovery is deterministic; tampering with any field invalidates the signature.

(3) Content-addressed. Order identifiers are content hashes (orderHash = keccak256(processId ∥ structHash
The same commitment content always produces the same identifier; different content always produces different identifiers (collision-resistant under the assumed hash function). This makes the record unforgeable in a specific sense: no second commitment can claim the same identifier.

(4) Immutable. On a finalized block, the record cannot be modified or deleted. A party seeking to repudiate a commitment cannot purge the chain of the relevant log entries. This is a stronger property than what electronic records normally enjoy.

(5) Publicly verifiable. Anyone can read the chain and replay the verification logic. The evidence is not held by any single party; it is held by the network. Disputing parties therefore cannot disagree about what the record says—they can disagree only about what the record *means*, which is precisely the question adjudication is suited to.

(6) Contains commitment intent. The signed payload includes a content hash h of the off-chain agreement. The off-chain agreement itself can be anything the parties choose—a structured contract, a free-text PDF, a schema-typed JSON document. The on-chain record does not interpret the agreement; it commits both parties to a specific document via the hash. A dispute over performance is then a dispute over the referenced document, with the on-chain hash establishing *which* document is at issue.

These six properties, combined, produce an artifact that is substantially *stronger* than the electronic records most courts already accept. Courts admit emails, electronic signatures, and database extracts as evidence subject to authentication challenges; the on-chain record satisfies many authentication requirements at the cryptographic level rather than at the custodial level.

5 Class A and Class B Records: A Taxonomy

The on-chain record's evidentiary use turns on a distinction that we have used inline in earlier sections and that deserves explicit treatment. We taxonomize the protocol's records into two classes distinguished by how they are produced and what they prove.

Class A: kernel byproducts. Class A records are emitted by the kernel itself as a byproduct of `commit` and `resolveProcess`. They include the commitment digest (the EIP-712 hash of the typed agreement struct), the bond deposits, the cumulative-value accumulator state, the order status transitions, and the resolution event with its complete payout vector. Class A records share four properties.

Discretion-free. The kernel emits Class A records without any participant choice; if the operation succeeded, the record is emitted in the form the kernel's transition function specifies. There is no off-chain choice about whether to record, what to record, or how to format the record.

Cryptographically authenticated. Each Class A record is signed by the parties who participated in the operation that produced it (EIP-712 signatures on commits; the buyer's transaction signature on resolutions). The cryptographic authentication is at the kernel level; an off-chain forum can verify the signatures without consulting any oracle or attestation service.

Append-only. Class A records cannot be modified after emission. The on-chain ledger's immutability properties apply fully; a Class A record at block b is the same record at block $b + N$ for any N.

Schema-typed at the kernel level. The structure of a Class A record is determined by the kernel's typed events; the fields are known in advance and machine-readable.

The four properties together make Class A records *strong evidence*: an off-chain forum can rely on the record's existence and content with high confidence and minimal expert testimony. The MLETR functional-equivalence analysis (Section 6) applies directly to Class A records, with the "reliable method" test discharged by the kernel's bytecode and the consensus rules of the underlying chain.

Class B: discretionary attestations. Class B records are produced by participants through the `AttestationCoordinator` contract under schema-validation discipline. They include delivery confirmations, GHG disclosures, handoff attestations, custody-of-seal events, lifecycle stage transitions, and the other schema-typed records that operational deployments produce. Class B records share four properties that contrast with Class A.

Discretionary. A participant chooses whether to file a Class B attestation, when to file it, and (within the schema's content space) what specifically to attest. The kernel does not require any specific Class B record to be filed for any specific process; whether a deployment requires a particular Class B record is determined by the off-chain agreement and the schema clauses committed at signing time.

Cryptographically authenticated, plus validator-gated. Class B records carry the participant's EIP-712 signature (authenticating who filed) plus the schema-validator's gate (authenticating that the content is well-formed under the registered schema). The combination produces stronger authentication than the participant's signature alone, but the content itself is participant-asserted and may be contestable on substantive grounds (was the delivery actually made? did the GHG disclosure reflect actual emissions?). The validator-gate's role in the protocol-tier integrity surface is part of schema design and is treated in that literature.

Append-only. Class B records, like Class A, cannot be modified after emission. Participants who attest incorrectly cannot retract; they can only file additional attestations (typically under a different schema or stage) that contextualize or supersede the earlier one.

Schema-typed at the protocol level. The structure of a Class B record is determined by the registered schema's specification (Layer A in the four-layer verification stack), ABI-encoded for on-chain validation (Layer C), and rendered for off-chain inspection through the runtime tier's schema-aware modules.

The four properties together make Class B records *useful but contestable evidence*: an off-chain forum can rely on the record's existence (cryptographic), its participant (cryptographic), and its formal correctness under the schema (validator-gated), but the substantive accuracy of what the record asserts (the delivery did happen as the seller stated; the GHG disclosure does reflect reality) raises questions the audit-evidence framework of AICPA [2020] treats across the assertion categories of existence (did the recorded event happen?) and accuracy (was it recorded at the right amount, timing, and characterization?). For Class B records the existence and accuracy assertions are both at stake; the wider seven-assertion AU-C 500 framework (existence, completeness, accuracy/valuation, cutoff, classification, occurrence, rights/obligations) supplies the language an audit-aware reader will reach for. We use "existence-and-accuracy" as shorthand for that broader frame.

The MLETR fit by class. The MLETR analysis applies most directly to Class A records, which carry the kernel's structural properties (singularity through the cumulative-value accumulator; control through buyer-dominance; integrity through the chain's consensus) and discharge MLETR Articles 10–11's control and singularity tests in their own right. Class B records do not generally meet the MLETR test on their own, because they are not transferable records in the Article-10 sense: a delivery confirmation, a GHG disclosure, or a custody-of-seal event is not an instrument of title that participants treat as functionally equivalent to a negotiable paper. Class B records are evidentiary artifacts *about* the underlying transferable record (the commitment), not separate transferable records inheriting the Article 10–11 status. Their evidentiary weight in any forum that imports MLETR-style functional-equivalence reasoning is as corroboration to the Class A commitment, conditional on the schema-validation surface's correctness (a Class B record under a schemaId without a registered validator does not get filed in the first place; the validator-gate at the `AttestationCoordinator` reverts). The schema-design discipline of Paper N is therefore part of the corroborative weight Class B records carry, not part of an MLETR transferable-record analysis applied to them directly.

Why the taxonomy matters for forum strategy. A litigant taking the bonded record to a forum should expect the forum's treatment of Class A and Class B records to differ in detail even if both are admissible.

For Class A, the forum's analysis typically reduces to: authentication (signatures verify; chain matches; expert if needed), then content interpretation (what does the recorded state mean for the parties' claims). The authentication step is nearly mechanical; the content-interpretation step is the work.

For Class B, the forum's analysis typically expands to include: authentication (as for Class A), schema-validity (was the record filed under a registered schemaId; did it pass the validator), then content interpretation *plus substantive truth-of-content* (does the attestation reflect the off-chain reality it asserts). The substantive truth-of-content question engages the AU-C 500 existence and accuracy assertions at the per-attestation granularity.

The difference matters operationally because a deployment that relies on Class B records to make load-bearing factual claims (e.g., a delivery attestation that determines whether the buyer's resolution is justified) is exposed to a different litigation profile than one that relies on Class A records (e.g., the resolution event itself, which is undeniable on chain). Most deployments use both classes; understanding the litigation profile of each is part of the runtime tier's design discipline.

6 Legal Fit: Existing Electronic-Evidence Frameworks

We sketch how the six properties above interact with four major electronic-evidence frameworks. The treatment is illustrative; a full doctrinal analysis is jurisdiction-by-jurisdiction.

eIDAS (EU Regulation 910/2014). eIDAS distinguishes electronic signatures, advanced electronic signatures (AdES), and qualified electronic signatures (QES). EIP-712 signatures meet the AdES technical requirements (uniquely linked to the signatory; capable of identifying the signatory; created using means under the signatory's sole control; linked to the signed data such that any subsequent change is detectable). They do not, by themselves, meet the QES requirements (which include a qualified certificate from a trust service provider). However, the AdES status is sufficient for most contractual purposes, and the eIDAS framework explicitly provides that the legal effect of an electronic signature shall not be denied solely because it is in electronic form. eIDAS also defines qualified electronic time stamps; block timestamps on chains with public consensus rules satisfy the technical requirements but typically not the trust-service-provider designation. The practical effect: the on-chain record is admissible electronic evidence under eIDAS, with weight determined by the court's contextual assessment.

UCC §1-201 and the Federal Rules of Evidence (US). The Uniform Commercial Code defines "record" as "information that is inscribed on a tangible medium or that is stored in an electronic or other medium and is retrievable in perceivable form." On-chain log entries satisfy this definition without ambiguity. UCC §1-201 also defines "signature" inclusively ("any symbol executed or adopted with present intention to adopt or accept a writing"); EIP-712 signatures, being intentionally executed by the signer's private key on the typed commitment, qualify. The Uniform Electronic Transactions Act (UETA) and the federal E-SIGN Act extend the same principle to contracts more broadly.

For evidentiary admission specifically, the operative provisions are Federal Rules of Evidence 901(b)(9), 902(13), and 902(14), together with the *Daubert* standard for any expert testimony required to authenticate the cryptographic process. *FRE 901(b)(9)* permits authentication of evidence by showing that "a process or system" produces an accurate result; on-chain records authenticate under this rule on the strength of the consensus protocol that produced them combined with public availability of the bytecode that processed the inputs. *FRE 902(13)* (added 2017) permits self-authentication of records "generated by an electronic process or system," and *FRE 902(14)* permits self-authentication of data copied from electronic devices through a process producing a digital identifier; both rules were drafted with electronic forensic evidence in mind and apply directly to on-chain records, where the "digital identifier" is the cryptographic hash of the record itself [FRE Legislative History, 2017]. The combined effect is that the on-chain record can be self-authenticated under 902(13)–(14) in many cases, with 901(b)(9) and Daubert available as the cross-examination route where an opposing party challenges the authentication. The practical effect: on-chain commitments are records and signatures under US commercial law, admissible under the FRE through the established electronic-evidence pathway, and authenticatable through *Daubert*-qualified expert testimony where authentication is challenged.

Singapore Electronic Transactions Act (Cap. 88) and MLETR. Singapore's ETA closely follows the UNCITRAL Model Law and provides functional equivalents for "writing", "signature", and "original" in electronic form. The ETA was amended in 2021 to recognize electronic transferable records, implementing the UNCITRAL Model Law on Electronic Transferable Records (MLETR, 2017) and extending the ETA's scope to instruments of title. On-chain records satisfy the writing and signature requirements directly; the transferable-record provisions warrant closer treatment because the doctrinal question they raise is the one most directly relevant to the process-tree structure of FIGARO.

MLETR Article 10 requires functional equivalence with a paper-based transferable document on three dimensions: (i) the information contained in the electronic record is accessible for subsequent reference; (ii) the record uses a reliable method (a) to identify the record as the electronic transferable record, (b) to render the record capable of being subject to control from its creation until it ceases to have effect, and (c) to retain the integrity of the record. Article 11 requires that "control" over the record be functionally equivalent to possession of a paper instrument: a single person identifiable as the controller, with exclusive capacity to transfer that control, and the transfer itself recorded reliably. The control and singularity tests are the load-bearing features.

FIGARO's on-chain commitment satisfies MLETR Articles 10–11 in the relevant senses. The orderHash uniquely identifies the transferable record (Article 10(2)(a)). The bond mechanics give the bonded party exclusive capacity to act on the record until resolution (Article 11(1) singularity); the cumulative-value accumulator ensures that the record cannot be duplicated or forked without the chain rejecting the duplicate (Article 10(2)(b) non-duplication). The integrity of the record is enforced cryptographically through the chain's consensus and through EIP-712 signature verification (Article 10(2)(c)). Where the right to receive performance moves along a process tree, each sub-order is itself a transferable record under the same analysis, and the on-chain settlement records the transfer with the reliable method MLETR contemplates. The Goode and Gullifer [2017] treatment of transferable records gives the standard doctrinal framework against which this analysis is run; the on-chain settlement satisfies the framework's tests without controversy.

For *Class-A* artifacts (those produced as kernel byproducts: commitment digests, settlement events, the cumulative-value accumulator) the MLETR fit is direct. For *Class-B* artifacts (discretionary attestations, schema-validated content) additional functional-equivalence analysis applies, since the MLETR analysis above turned on the kernel's structural properties rather than on attestation semantics; we do not extend the analysis to Class-B here.

UNCITRAL Model Law on Electronic Commerce. Adopted in 1996 and supplemented by the 2017 Model Law on Electronic Transferable Records, the UNCITRAL framework articulates the functional-equivalence principle that underlies most modern electronic-evidence regimes worldwide. The Model Law's three core requirements—accessibility for subsequent reference, integrity of the information, and identifiability of the originator—are all satisfied by on-chain commitments. The practical effect: jurisdictions that have adopted UNCITRAL-derived electronic-transactions law (over 80 as of 2024) provide an established framework for treating on-chain commitments as admissible electronic evidence.

The summary picture. The on-chain record is admissible electronic evidence in every jurisdiction whose electronic-evidence law derives from the UNCITRAL framework or implements equivalent functional principles. What varies across jurisdictions is the weight courts assign to the evidence (a question of judicial practice rather than admissibility) and the further procedural requirements for authenticating the chain itself (typically satisfied by expert testimony or by the court taking judicial notice of well-known chains). No jurisdiction we are aware of categorically excludes on-chain records from evidentiary consideration.

7 Substantive Defenses and Post-Settlement Restitution

A reader sympathetic to the argument so far will raise the sharpest question this paper has to answer. Suppose the on-chain settlement is unimpeachable as evidence; suppose every framework above admits it. What happens when an external court, applying the substantive contract law of its jurisdiction, finds that the underlying agreement was procured under duress, formed under fundamental mistake, frustrated by impossibility, voided by illegality, or unconscionable in its terms? The on-chain settlement has, by hypothesis, already executed: bonds are released, payouts distributed, and the kernel's state shows the process resolved. The question is what the court can do with that fact and whether the substitution between settlement (on-chain) and adjudication (off-chain) the present paper has defended is consistent with the substantive-defense jurisprudence the dominant tradition has developed over centuries.

We treat the question in three registers: the doctrinal inventory, the substantive-defenses analysis, and the restitutionary mechanics by which a court order to unwind interacts with an already-discharged kernel state.

Doctrinal inventory. The defenses that void or rescind a contract *ab initio*, or that excuse subsequent performance, are doctrinally settled in the major commercial jurisdictions. Common-law systems recognize: *duress* (consent obtained by improper threat); *undue influence* (consent obtained through abuse of a relationship of trust); *misrepresentation* (consent obtained by false statements about material facts); *fundamental mistake* (consent given under a shared mistake about a fact essential to the agreement); *unconscionability* (terms so one-sided that enforcement shocks the conscience of the court); *frustration of purpose* (subsequent events render performance fundamentally different from what was undertaken); *impossibility / impracticability* (subsequent events render performance physically or practically impossible); *illegality* (the agreement requires conduct prohibited by law); *public policy* (enforcement would violate public policy independent of any specific statute). Civil-law systems recognize closely parallel doctrines under different names (*vices du consentement*, *erreur*, *circonstances exceptionnelles*, etc.). The doctrines are substantive: they apply to *the agreement* the bonded commitment records, not to the bonded commitment as a technological artifact.

Substantive-defenses analysis. The bonded primitive is silent on every doctrine in the inventory. The kernel does not know whether consent was procured under duress, whether parties were mistaken about a material fact, whether the underlying transaction was illegal, or whether performance has been frustrated by external circumstance. The kernel records what was signed; it makes no representation about the conditions under which signing occurred or the legality of what was signed for. This is by design — the kernel cannot make substantive-defense determinations without doing the kind of discretionary work the no-escape-hatches design principle of the mechanism paper rules out — and it is by design also a *feature* for the present paper's purpose, because it leaves the substantive-defense apparatus exactly where the dominant tradition has placed it: in courts applying their own constitutional and doctrinal apparatus to the agreement recorded in the bonded commitment.

The leading case-law authority on smart-contract substantive defense is *Quoine Pte Ltd v B2C2 Ltd* [Quoine v B2C2 , 2019, 2020]. The Singapore International Commercial Court (and on appeal the Court of Appeal) held that a smart-contract execution can be subject to the equitable doctrine of unilateral mistake, and that a party who knows or ought to know of a counterparty's mistake at the moment a smart contract executes can be required to make restitution. The Quoine line is the authoritative answer that substantive-defense doctrines apply to smart-contract executions in the jurisdictions whose courts are willing to recognize them, and the bonded primitive is no exception.

Restitutionary mechanics. The mechanics are where the question becomes operationally sharp. Three sub-cases.

Case 1: court orders specific re-transfer off-chain. Where the court determines that one party should not have received the on-chain payout (e.g., because the underlying agreement was void for fundamental mistake), the standard remedy is a restitutionary order requiring that party to transfer the received value back to the other off-chain. This remedy operates on the parties' real-world capacity to comply, not on the kernel state. The kernel records show the original settlement unchanged; the off-chain transfer is recorded separately, typically through conventional financial rails or through a fresh on-chain transaction. The kernel state is, in this arrangement, evidentiary input to the dispute and operational substrate for the off-chain remedy; it is not itself the locus of the unwinding. This is the most common form of post-settlement restitution and the form for which the existing legal apparatus is well-equipped.

Case 2: court orders replacement-bonded counter-process. Where ongoing performance is required and a counter-flow is appropriate, the parties (or the court via an order on the parties) can constitute a fresh Figaro process that flows value in the corrective direction. This uses the bonded primitive as the implementation of the court's remedy rather than as an obstacle to it. The original process remains resolved; the new process is independent and bonded under the same kernel.

Case 3: court orders an arrangement the kernel cannot support. Where the court's appropriate remedy cannot be implemented within the kernel's bilateral primitive (e.g., because it requires the kind of discretionary administrative authority the kernel forbids), the remedy operates on the parties' off-chain assets and obligations through conventional mechanisms (asset seizure, contempt sanctions, criminal liability for non-compliance with civil orders). The kernel does not obstruct this remedy; it does not implement it either. This is consistent with the general principle that the substrate is sub-political, not anti-political (Section 10 returns to this).

Implication for Paper A's escape-hatch result. The mechanism paper's *escape-hatch weakness* theorem (Theorem 4.7 of Paper A) and the carve-out in its Remark 4.8 deserve a more precise characterization than either text gives. The theorem rules out unilateral exit paths internal to the mechanism — timeouts, arbitrators, governance votes — because those would weaken the Nash equilibrium. Remark 4.8 carves out external legal forums applying duress, frustration, and similar substantive doctrines, on the ground that those forums are constrained by their own institutional bond structures and operate on the bonded commitment as evidentiary input.

The carve-out is sound on closer analysis but for a different reason than Remark 4.8 suggests. The relevant fact is not that external forums are themselves bonded; it is that the substantive-defense jurisprudence operates on the agreement *as recorded* after the kernel has settled, not on the kernel itself, and produces remedies that reach the parties through their conventional legal exposure rather than through the bonded escrow. The kernel's equilibrium is preserved because no remedy can reach into the discharged escrow; the substantive defenses operate downstream of settlement. A participant deciding whether to bond does so with the understanding that the bonded commitment is enforceable as between the parties and may, separately, be tested in court under conventional contract doctrines on the same terms as any other contract. Both the on-chain enforcement and the off-chain doctrinal review apply; they are not in competition because they reach different aspects of the same arrangement. The bonded primitive does not displace contract law's substantive doctrines; it produces a settlement under which those doctrines continue to apply in the conventional way.

8 A Worked Example: New York Forum, Cross-Border Process

A worked example sharpens the abstract analysis. Consider a process tree with a buyer in New York, a primary seller in Singapore, and a sub-seller in Argentina, settled in `USDC` on Ethereum L1. The off-chain agreement specifies New York Supreme Court as the forum and New York law as the governing law (a common contractual arrangement for cross-border commerce involving a US counterparty). A dispute arises: the primary seller claims the buyer never resolved the process despite delivery; the buyer claims the goods did not conform to the agreed specification.

The bonded primitive's contribution to the dispute is direct. The on-chain record establishes: the commitment hash matching the off-chain agreement (UCC §1-201 record; FRE 902(13) self-authenticating); the buyer's and seller's signatures on the EIP-712 typed commitment (UCC §1-201 signature; AdES under eIDAS; signature under E-SIGN/UETA); the cumulative-value accumulator's monotonic state through every commit and resolve operation; and the absence (or presence) of a `resolveProcess` transaction. The seller's claim — that the buyer never resolved — is determinable from the chain state directly. The buyer's claim — that the goods did not conform — is not determinable from the chain state and requires off-chain evidence: delivery records, photographs, laboratory analyses, and (where the process used the FIGARO-fulfilment schema) the schema-validated attestations that may already be on chain.

The New York court, applying New York contract law (UCC Article 2 for the goods, NY CPLR for the procedural framework, NY GBL §75-a for electronic signatures), proceeds as follows. Admissibility of the on-chain record is established under the FRE pathway above (902(13) self-authentication; 901(b)(9) process-or-system authentication; *Daubert*-qualified expert if needed). The substantive question — did the goods conform? — is decided on the off-chain evidence under ordinary contract doctrine. The remedy, if the court finds for the buyer, is restitutionary: an order requiring the seller to repay the unconformed portion of the payment. The order operates on the seller's off-chain assets through standard enforcement mechanisms (US–Singapore reciprocity, New York Convention, asset attachment in either jurisdiction).

The bonded settlement does not interfere with any of this. It provides the evidentiary record and the bilateral-enforcement backstop that prevents the seller from walking away with the funds before the dispute is heard — the usual gap that cross-border commerce has had to manage through letters of credit, escrow agents, and intermediary banks. The bonded primitive supplies that function more cleanly, leaving the substantive doctrine of the New York court untouched. This is the architecture the present paper has been describing: the primitive replaces the enforcement gap that conventional commerce has had to fill with intermediaries; it does not replace the substantive doctrine that the court applies, and it is no less well-positioned in cross-border settings than in domestic ones because the evidentiary properties are the same in either case.

9 Jurisdictional Implications

The decoupling of settlement from adjudication has substantive jurisdictional consequences.

Forum is contractual. The on-chain commitment includes an off-chain agreement hash h. That off-chain agreement can—and typically should—include a forum-selection clause, a choice-of-law clause, and an arbitration clause if the parties prefer arbitration to court. The on-chain record proves what was agreed; the off-chain agreement specifies where and under what law the agreement is adjudicated. Parties who omit a forum clause default to whatever private-international-law analysis the seised forum applies; this is a known regime, not a novel problem.

Forum-shopping is bounded. A party seeking to forum-shop is bounded by (a) the forum-selection clause in the off-chain agreement, (b) the seising forum's own jurisdictional rules, and (c) the recognition-and-enforcement treaties between the seising forum and any forum where the counterparty's assets reside. None of these constraints is weakened by the on-chain settlement; they all operate on the off-chain agreement and the parties' real-world circumstances.

Evidence is jurisdiction-displaced. The integrity of the on-chain evidence does not depend on the seising forum's recognition of any specific cryptographic standard. The signatures verify, or they do not; the chain hashes match, or they do not. A forum that doubts the cryptographic analysis can engage an expert; a forum that does not is in the position of any forum facing forensic evidence.

Cross-border processes. A process tree can involve parties in multiple jurisdictions simultaneously. The on-chain settlement is jurisdiction-blind: the bond mechanics work identically regardless of where the parties reside. Off-chain disputes that arise from such processes are handled by the same private-international-law machinery that already handles cross-border disputes in conventional commerce—typically with the on-chain record providing substantially more evidentiary clarity than the parties would have produced under conventional bookkeeping.

10 Why Not On-Chain Adjudication

We have argued the positive case for off-chain adjudication. We now argue the negative case against on-chain adjudication. Several patterns occupy this space; we treat each briefly.

Schelling-point juror selection (Kleros). A token-weighted juror panel votes on the disputed question; the plurality wins; jurors who voted with the plurality are rewarded, jurors who voted against are slashed. The mechanism is elegant in its abstraction but has three problems for serious adjudication. First, the juror panel is not selected for jurisdictional fit—a Kleros juror in jurisdiction A may know nothing of jurisdiction B's law that the parties agreed should govern. Second, the juror's incentive is to vote with the perceived plurality, not to vote for the legally correct outcome—a Schelling point on plurality is structurally distinct from a Schelling point on truth. Third, the slashing mechanism imports a new dispute (was the slashing fair?) that has no terminal forum on chain.

DAO-governed arbitration (Aragon Court). Aragon Court is structurally similar to Kleros, with the juror-selection layer integrated with DAO governance. This inherits all of Kleros's problems plus the additional problem that DAO voting power is plutocratic in proportion to token holdings; the wealthy juror has structurally more votes than the poor juror. Whether or not one views this as a desirable property of arbitration is a normative question; that it differs from the conventional "one juror, one vote" premise of jury trials is a doctrinal question that has not been settled.

Optimistic challenge windows. Optimistic mechanisms [Optimism, 2021, UMA, 2020] avoid the juror problem by allowing assertions to stand unless challenged within a fixed window. This works for settlement-layer state machines but does not work for adjudication, because adjudication is precisely the activity of assessing an assertion that has been challenged. An optimistic "adjudication" is just an assertion with a deadline—it has no mechanism for resolving the underlying question, only for resolving the question of who blinked first.

The structural objection. All on-chain adjudication mechanisms share the same structural flaw. The kernel mechanism of the mechanism paper establishes that any unilateral exit from the bonded state requires either a third-party decision (which introduces an unbonded actor whose incentives are not constrained by the bond structure) or a payoff modification that weakens the Nash equilibrium. On-chain adjudication is the third-party-decision case: the juror, the DAO voter, the challenger is the unbonded actor whose decision determines the outcome. Whatever the mechanism designer does to constrain that actor (token slashing, Schelling incentives, reputation), the actor is still unbonded and the kernel's self-enforcing property is broken.

The off-chain alternative. Off-chain adjudication by an existing legal forum has the opposite structure. The forum's incentives are constrained by its institutional context—judges face appellate review, arbitrators face institutional reputation, both face professional sanction. The forum is not unbonded in the kernel sense; it operates inside a different bond structure (its institutional authority). The off-chain evidence record is therefore consumed by an actor whose incentives are already constrained by a regime that has been refined for centuries. The protocol does not re-implement that regime; it provides input to it.

Composition with Kleros as forum, not as kernel adjudicator. The objection above is to on-chain adjudication *at the kernel layer*. It is not an objection to Kleros, or to any other adjudication mechanism, when the parties' agreement designates it as the forum. The kernel of FIGARO is forum-agnostic: it does not adjudicate; it produces an evidence record. If two parties sign an off-chain agreement whose arbitration clause names Kleros as the dispute forum, the evidence record produced by their FIGARO commitments is exported to Kleros and the dispute is adjudicated there. From the kernel's perspective Kleros is the off-chain forum the parties chose, in the same sense that the Singapore International Arbitration Centre or the New York Supreme Court can be the off-chain forum the parties choose. The structural critique above is not an endorsement of some forums over others; any forum the parties name in their agreement is their forum to choose, and FIGARO composes with it by exporting evidence to it.

Engagement with the legal-academic literature on Kleros. A small but real legal-academic literature has developed around Kleros and its peers. Ortolani [2019] is the principal engagement from the international-arbitration side, raising the recognition-and-enforcement question under the New York Convention [New York Convention, 1958]: can a Kleros decision be enforced abroad as an arbitral award, and on what doctrinal basis? Kaal [2020] reviews the broader landscape of decentralized dispute resolution. De Filippi and Wright [2018] situates Kleros and similar mechanisms within the lex cryptographia frame. Rabinovich-Einy and Katsh [2017] place them in the longer trajectory of online dispute resolution. The literature is not yet large but it is real, and its general posture is compatible with this paper's: most of the legal-academic engagement with Kleros expresses concern about adjudicative-quality and recognition issues, not about the existence of the mechanism.

Our committed view on the New York Convention question. The Kleros recognition question deserves a committed answer rather than a deferral. Our reading of Article I of the New York Convention — which defines its scope to "arbitral awards made in the territory of a State other than the State where the recognition and enforcement of such awards are sought" — is that a Kleros decision is capable of falling within the Convention's definition of an arbitral award where the parties' arbitration agreement so designates, subject to Article V's grounds for refusing recognition (most relevantly the public-policy ground of Article V(2)(b) and the "proper law of the arbitration agreement" analysis under Article V(1)(a)). The recognition is not automatic — a court asked to enforce a Kleros decision can refuse on Article V grounds, and the substantive-defense doctrines of Section 7 apply on the merits as they would for any arbitral award. The recognition is also jurisdictionally sensitive: Convention-implementing legislation varies, and common-law and civil-law jurisdictions have different default postures toward unconventional arbitration arrangements. Our view is therefore that Kleros decisions *can* be arbitral awards under the Convention, conditional on competent drafting of the arbitration agreement and on the seising court's willingness to read the Convention's definition of an "arbitral tribunal" broadly enough to include the juror-selection mechanism Kleros operates. The contrary view [on which see Ortolani, 2019] is defensible and turns on the same Article I and Article V considerations. We do not treat the question as settled, but we do treat it as takable, and we have written the architecture to be neutral on the choice of forum so that parties whose recognition theory favors a state-court forum can choose one and parties whose theory accommodates a Kleros forum can choose Kleros, with FIGARO's evidence record exported to either.

The recognition-and-enforcement question raised by Ortolani and others is in our reading the binding legal-doctrinal question, and the layered architecture of Section 2 mitigates it: the on-chain record is admissible electronic evidence under the frameworks of Section 6 regardless of how the forum that consumes the evidence is constituted, and forum-clause discipline allows the parties to name a forum (Kleros, SIAC, court) whose decisions they have a clean recognition theory for in the jurisdiction of intended enforcement.

11 Composition with Dispute Systems

The previous section makes the negative case (why on-chain adjudication breaks the equilibrium); a constructive treatment is also owed to the practitioner who arrives with the question "which forum should I name in my agreement, and how do I compose with it." We catalogue the four classes of forum the protocol composes with in practice and treat each as a composition pattern.

11.1 State courts (conventional civil and commercial courts)

Conventional state courts are the highest-recognition forum available and the appropriate destination for disputes whose substantive resolution requires the full apparatus of contract law (duress, mistake, frustration, illegality, public policy — the doctrines treated in Section 7). The composition pattern is direct.

Forum clause. The off-chain agreement names a specific state court (e.g., New York Supreme Court, Singapore International Commercial Court, English High Court) and a governing law (e.g., New York law, Singapore law, English law). The clause is a standard contractual provision; the bonded primitive does not modify the law of forum selection.

Evidence path. The bonded record is admitted under the relevant evidentiary framework (Section 6: FRE 901(b)(9) / 902(13)–(14) for US courts, eIDAS AdES recognition for EU courts, Singapore ETA / MLETR for Singapore courts, analogous frameworks elsewhere). Authentication challenges are typically minimal because the cryptographic properties are self-authenticating; substantive challenges go to the merits in the ordinary way.

Recognition and enforcement. A judgment from a recognized state court is enforceable through the standard private-international-law machinery: bilateral and multilateral recognition treaties, the Hague Convention on Recognition and Enforcement of Foreign Judgments (where in force), and the forum's own domestic enforcement powers. Enforcement against on-chain assets requires a court with jurisdiction over the asset-holding party rather than over the bonded primitive itself (which has no jurisdiction in the relevant sense; the kernel is not a party).

When state courts are appropriate. Disputes involving substantive contract defenses, public-policy questions, sanctions enforcement, criminal conduct adjacent to the bonded commerce, regulatory enforcement, or any matter where the parties need the recognition properties of a state court's judgment.

11.2 International commercial arbitration centers

International commercial arbitration centers (the Singapore International Arbitration Centre / SIAC, the International Chamber of Commerce / ICC, the London Court of International Arbitration / LCIA, the Hong Kong International Arbitration Centre / HKIAC, and analogous institutions) are the appropriate destination for cross-border disputes whose parties prefer arbitration to court litigation. The composition pattern is similar to state courts but uses the New York Convention's recognition machinery rather than bilateral judgment-recognition treaties.

Forum clause. The off-chain agreement names the arbitration center, specifies the seat of arbitration, the language of the proceeding, and the institution's procedural rules (e.g., "arbitration administered by SIAC under its 2025 Rules; seat in Singapore; English language; three arbitrators appointed under Article 14").

Evidence path. Arbitration centers admit electronic evidence broadly and typically have flexible authentication standards; the bonded record satisfies the standards straightforwardly.

Recognition and enforcement. The New York Convention [New York Convention, 1958] is the canonical recognition framework for arbitral awards; 173 state parties as of 2024. An award rendered under SIAC / ICC / LCIA rules is enforceable in any New York Convention state subject to the Article V grounds for refusal (public policy, due process, etc.). The bonded primitive composes with this enforcement chain natively because the underlying agreement is a recognized arbitration agreement under Article II.

When arbitration centers are appropriate. Cross-border disputes where the parties prefer institutional arbitration's combination of expertise, procedural flexibility, and international enforceability over the conventional-court alternative; high-value disputes where the parties want arbitrators with sector-specific expertise.

11.3 Online dispute resolution (ODR) platforms

ODR platforms (eBay's Resolution Center as the volume archetype, Modria, smartsettle, and the broader online-dispute-resolution literature [Rabinovich-Einy and Katsh, 2017]) are the appropriate destination for high-volume, low-value disputes where the cost of conventional adjudication exceeds the disputed amount. The composition pattern is structurally distinct from courts and arbitration in that the ODR platform's recognition properties are typically weaker but its operational cost is much lower.

Forum clause. The off-chain agreement names the ODR platform and accepts its terms-of-service for dispute handling. The platform's binding effect is contractual rather than treaty-based.

Evidence path. ODR platforms typically accept the bonded record's evidentiary properties without specialized authentication; the platform's internal review uses the record as the dispositive content for most operational disputes (delivery confirmation, GHG disclosure verification, handoff attestation review).

Recognition and enforcement. ODR decisions are typically enforced by the platform itself (e.g., a refund or replacement ordered by the platform's review process) rather than through external state machinery. A practitioner pattern of pairing an ODR clause with an arbitration clause that treats unappealed ODR decisions as final and arbitration-recognized is structurally available but doctrinally delicate: the arbitration clause must be drafted to constitute the ODR review as the arbitral process (or as a binding precursor with a separate arbitral-confirmation step), and the resulting award's recognition under the New York Convention turns on whether the ODR proceeding satisfies Article V(1)(b) due-process requirements. Many ODR platforms' procedures (no live hearing, no document discovery, opaque review) fail Article V(1)(b) on standard readings, so the hybrid is a careful drafting matter rather than an off-the-shelf composition.

When ODR is appropriate. Routine commercial disputes at operational scale where the dispute volume rules out court/arbitration economics; deployments where the parties prefer fast resolution to definitive recognition.

11.4 Schelling-point juror systems (Kleros and analogues)

Kleros and analogous Schelling-point juror systems [Kleros, 2019] occupy a distinctive position. They are not state courts, not traditional arbitration centers, and not conventional ODR platforms. They are mechanism-based forums where a Schelling-point juror selection produces decisions through coordination on the focal answer. The protocol composes with them as the parties' chosen forum, but the recognition and enforcement properties are the most contested of the four classes.

Forum clause. The off-chain agreement names Kleros (or analogue), specifies the court within Kleros (the contracting parties choose from the registered courts in the Kleros system), and specifies the procedural rules (juror count, evidence-period duration, appeal availability).

Evidence path. The bonded record is exported to Kleros through the parties' filing of the dispute; jurors evaluate the evidence and vote.

Recognition and enforcement. Section 10 above developed the committed view: a Kleros decision is *capable* of falling within the New York Convention's definition of an arbitral award where the parties' arbitration agreement so designates, subject to Article V grounds for refusing recognition. Ortolani [2019] raises three specific objections that the committed view must engage. First, NYC Article I requires an identifiable "tribunal", and Kleros jurors are anonymized, rotating, and token-incentivized rather than arbitrators in the institutional sense the Convention has historically contemplated. Our committed view contests this objection on the ground that the Convention's tribunal definition is functional rather than institutional — the relevant question is whether a proceeding produces a decision on the merits between the parties, not whether it does so through institutionally-credentialed arbitrators. Second, Article V(1)(d) requires that "the composition of the arbitral authority" was in accordance with the parties' agreement, and algorithmic juror selection raises the question of whether "composition" has been agreed in the relevant sense. Our committed view contests this objection by reading the parties' acceptance of the Kleros court's procedural rules (which the parties opt into when naming the court) as constituting the required agreement on composition. Third, Article V(2)(b) permits refusal of recognition on public-policy grounds, and many civil-law jurisdictions treat unreasoned awards as presumptively contrary to public policy — Kleros decisions are vote tallies, not reasoned awards. *Our committed view concedes this objection in civil-law jurisdictions whose public policy requires reasoned awards.* Recognition of Kleros decisions in such jurisdictions is materially harder than the generic analysis suggests, and parties enforcing in those jurisdictions should pair the Kleros clause with a written opinion supplement or choose a different forum. The committed view is therefore: takable on Article I and Article V(1)(d) on the functional readings above; not takable on Article V(2)(b) in civil-law public-policy jurisdictions whose reasoned-award requirement applies.

When Kleros is appropriate. Disputes whose parties prefer the Schelling-point juror mechanism's combination of neutrality (no state authority), speed (faster than conventional arbitration), and the token-economic incentive structure for juror participation, over the recognition properties of conventional forums; operational deployments where the bond posture itself produces strong enough enforcement that the forum's recognition is needed only in a small minority of cases. Unsuitable for deployments whose intended enforcement jurisdictions include civil-law public-policy regimes that require reasoned awards.

11.5 Composition discipline across the four

Two structural points apply across all four forum classes.

Forum-clause discipline is the parties' choice and the parties' responsibility. The bonded primitive does not recommend a forum; it does not prefer state courts to arbitration centers, ODR to Kleros, or any of the four to any other. The forum is named by the parties' off-chain agreement, and the parties bear the consequences of their forum choice (recognition properties, enforcement chain, procedural-rule adoption, operational cost). The architecture's role is to produce evidence the chosen forum can use, not to constrain the choice.

Recognition theory is upstream of forum choice. The parties should name a forum whose decisions they have a clean recognition theory for in the jurisdiction of intended enforcement. A US-based seller and an EU-based buyer who name SIAC arbitration with Singapore seat have a clean recognition theory through the New York Convention. The same parties naming an ODR platform without arbitration backing have no clean recognition theory if the ODR decision is contested. The same parties naming a state court whose judgments are not recognized in the counterparty's jurisdiction have a recognition gap that the bonded primitive cannot close.

The bonded primitive shifts the balance toward weaker forums. Because the bonded posture itself enforces most disputes at Layers 1 and 2, the residual that reaches Layer 3 is small (Section 3). A smaller residual volume can sometimes justify a weaker forum than the deployment would have chosen under conventional commerce: a deployment that would have insisted on SIAC arbitration in conventional commerce might choose an ODR platform under the bonded primitive because the residual volume reaching Layer 3 is small enough that ODR's weaker recognition is acceptable. This shift is not the same as saying the architecture replaces stronger forums; it is saying the architecture changes the cost-benefit calculation of which forum is appropriate for what residual.

12 Conclusion

We have argued for a layered approach to smart-contract dispute resolution: settlement on chain, adjudication off chain, with a deliberately designed evidentiary record that bridges the two. The architecture inherits the strengths of each layer (chain determinism for settlement, legal-system contextual judgment for adjudication) without forcing either to imitate the other. It also resolves the structural objection to on-chain adjudication: the unbonded juror does not need to exist because the legally constituted forum already does.

The contribution is modest but, we believe, important. The debate over smart-contract dispute resolution has been dominated by the assumption that on-chain settlement implies on-chain adjudication. The decoupling we propose is a different design posture, and it permits a substantially cleaner alignment with existing electronic-evidence law.

What remains for legal scholarship. Three questions are out of scope for this paper but warrant sustained legal-scholarly attention. First, the doctrinal analysis of on-chain commitments under specific jurisdictions' contract-formation, evidence, and electronic-signature laws is substantial and necessarily jurisdiction-by-jurisdiction. Second, the question of how courts should address chain-finality assumptions (what happens when the underlying chain reorganizes substantially after a contested transaction) requires technical-legal collaboration that has barely begun. Third, the question of whether and when the on-chain record should be treated as self-authenticating (in the US Federal Rules of Evidence sense, or its analogues in other common-law jurisdictions) is a question of judicial practice that will develop with case law.

What remains for protocol design. On the protocol side, the principal open question is how schema-typed attestations (implementation paper) interact with off-chain dispute proceedings. The attestation surface allows parties to record structured evidence (geo proofs, lifecycle stage transitions, GHG disclosures) on chain in addition to the bare commitment. How that structured evidence is treated by off-chain forums—whether as supplementary fact, as binding stipulation, or as expert testimony—is design space that has not yet been explored.

Acknowledgments

This argument owes a substantial debt to the Coalition of Automated Legal Applications (COALA), whose work over many years has established the framework within which questions of this kind can be productively asked. I thank colleagues there and at the Stanford CodeX Center for conversations that sharpened the distinction between settlement and adjudication that this paper develops.

References

Cuende, L. and Izquierdo, J. Aragon Network: A Decentralized Infrastructure for Value Exchange. Aragon White Paper, 2017.

De Filippi, P. and Wright, A. *Blockchain and the Law: The Rule of Code.* Harvard University Press, Cambridge, MA, 2018.

Bloemen, R., Logvinov, L., and Evans, J. EIP-712: Ethereum typed structured data hashing and signing. Ethereum Improvement Proposal, 2017. `https://eips.ethereum.org/EIPS/eip-712`

Kaal, W. A. Decentralized Dispute Resolution. University of St. Thomas (Minnesota) Legal Studies Research Paper No. 20-13, 2020.

Ast, F. and Lesaege, C. Kleros: A Protocol for a Decentralized Justice System. Kleros White Paper, 2019. `https://kleros.io/yellowpaper.pdf`

United Nations. *Convention on the Recognition and Enforcement of Foreign Arbitral Awards* (the New York Convention). 330 U.N.T.S. 38, 1958.

Optimism Foundation. Optimistic Rollups. Optimism Technical Documentation, 2021.

Ortolani, P. The Impact of Blockchain Technologies and Smart Contracts on Dispute Resolution: Arbitration and Court Litigation at the Crossroads. *Uniform Law Review*, 24(2):430–448, 2019.

Rabinovich-Einy, O. and Katsh, E. *Digital Justice: Technology and the Internet of Disputes*. Oxford University Press, New York, 2017.

Quoine Pte Ltd v B2C2 Ltd [2019] SGHC(I) 03 (Singapore International Commercial Court).

Quoine Pte Ltd v B2C2 Ltd [2020] SGCA(I) 02 (Singapore Court of Appeal).

US Judicial Conference, Advisory Committee on Evidence Rules. Report on Proposed Amendments to Federal Rules of Evidence (902(13) and 902(14)). Effective December 1, 2017.

Goode, R. and Gullifer, L. *Goode and Gullifer on Legal Problems of Credit and Security*, 6th edition. Sweet & Maxwell, London, 2017.

Yunus, M. *Banker to the Poor: Micro-Lending and the Battle Against World Poverty*. PublicAffairs, New York, 1999.

Morduch, J. The Microfinance Promise. *Journal of Economic Literature*, 37(4):1569–1614, 1999.

American Institute of Certified Public Accountants. *AU-C Section 500: Audit Evidence*. AICPA, Durham, NC, 2020.

UMA Project. The Optimistic Oracle. UMA Technical Documentation, 2020.